

Dual Representation and Kernel Methods

COMP0078: Supervised Learning (Continued)

Lecture Notes

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Revision notation: items marked with $\star$ are specifically listed in the revision notes file (`revision25-26-V1.pdf`). All content in this file is examinable.

1 Dual Representation

This section continues from the ridge regression setup. Recall the modified normal equations:

$$-2X^\top(\mathbf{y} - X\mathbf{w}) + 2\lambda\mathbf{w} = 0 \quad (1)$$

and the primal ridge regression solution:

$$\hat{\mathbf{w}}_\lambda = (X^\top X + \lambda I_n)^{-1} X^\top \mathbf{y} \quad (2)$$

as well as the dual form stating $\mathbf{w} = \sum_{i=1}^m \alpha_i \mathbf{x}_i$ with dual coefficients satisfying:

$$\boldsymbol{\alpha} = (XX^\top + \lambda I_m)^{-1} \mathbf{y} \quad (3)$$

and the dual prediction function:

$$f(\mathbf{x}) = \sum_{i=1}^m \alpha_i \mathbf{x}_i^\top \mathbf{x} \quad (*)$$

1.1 Derivation of the Dual (Step 1)

We rewrite the normal equation (1) as

$$\mathbf{w} = \frac{X^\top(\mathbf{y} - X\mathbf{w})}{\lambda} \quad (4)$$

Thus we have

$$\mathbf{w} = \sum_{i=1}^m \alpha_i \mathbf{x}_i \quad (5)$$

with

$$\alpha_i = \frac{y_i - \mathbf{w}^\top \mathbf{x}_i}{\lambda} \quad (6)$$

Consequently, we have that

$$\mathbf{w}^\top \mathbf{x} = \sum_{i=1}^m \alpha_i \mathbf{x}_i^\top \mathbf{x} \quad (7)$$

proving (*).

1.2 Derivation of the Dual (Step 2)

Plugging (5) into (6) we obtain

$$\alpha_i = \frac{y_i - \left(\sum_{j=1}^m \alpha_j \mathbf{x}_j \right)^\top \mathbf{x}_i}{\lambda} \quad (8)$$

Thus (defining $\delta_{ij} = 1$ if $i = j$ and 0 otherwise):

$$y_i = \left(\sum_{j=1}^m \alpha_j \mathbf{x}_j \right)^\top \mathbf{x}_i + \lambda \alpha_i \quad (9)$$

$$y_i = \sum_{j=1}^m (\alpha_j \mathbf{x}_j^\top \mathbf{x}_i + \alpha_j \lambda \delta_{ij}) \quad (10)$$

$$y_i = \sum_{j=1}^m (\mathbf{x}_j^\top \mathbf{x}_i + \lambda \delta_{ij}) \alpha_j \quad (11)$$

Hence $(XX^\top + \lambda I_m)\boldsymbol{\alpha} = \mathbf{y}$, from which (3) follows.

2 Computational Considerations

Remark 2.1 (Training Time Complexity*). Solving for $\mathbf{w}$ in the primal form (2) requires $O(mn^2 + n^3)$ operations, while solving for $\boldsymbol{\alpha}$ in the dual form (3) requires $O(nm^2 + m^3)$ operations.

If $m \ll n$ it is more efficient to use the dual representation.

Remark 2.2 (Testing Time Complexity*). Computing $f(\mathbf{x})$ on a test vector $\mathbf{x}$ in the primal form requires $O(n)$ operations, while the dual form (*) requires $O(mn)$ operations.

2.1 Sparse Representation

We can benefit even further in the dual representation if the inputs are sparse!

Example 2.3 (Sparse Inputs). Suppose each input $\mathbf{x} \in \mathbb{R}^n$ has most of its components equal to zero (e.g., consider images where most pixels are ‘black’ or text documents represented as ‘bag of words’).

- If k denotes the number of nonzero components of the input then computing $\mathbf{x}^\top \mathbf{t}$ requires at most $O(k)$ operations.

How do we do this?

- If $km \ll n$ (which implies $m, k \ll n$) the dual representation requires $O(km^2 + m^3)$ computations for training and $O(mk)$ for testing (cf. Remarks 2.1 and 2.2).

3 Kernels*

3.1 Computational Considerations Revisited

Again, if $m \ll N$ it is more efficient to work with the dual representation (Remark 2.1).

Remark 3.1 (Key Observation). In the dual representation (*) we don't need to know ϕ explicitly; we just need to know the inner product between any pair of feature vectors!

Example 3.2 (Second-Order Correlation Kernel). Consider the following feature map with second order correlations ($N = n^2$):

$$\phi(\mathbf{x}) = (x_1x_1, x_1x_2, \dots, x_nx_n)^\top \quad (12)$$

$$\langle \phi(\mathbf{x}), \phi(\mathbf{t}) \rangle = (x_1x_1, x_1x_2, \dots, x_nx_n)(t_1t_1, t_1t_2, \dots, t_nt_n)^\top \quad (13)$$

$$= x_1x_1t_1t_1 + x_1x_2t_1t_2 + \dots + x_nx_nt_nt_n \quad (14)$$

$$= (x_1t_1 + \dots + x_nt_n)(x_1t_1 + \dots + x_nt_n) \quad (15)$$

$$= (\mathbf{x}^\top \mathbf{t})^2 \quad (16)$$

Observe that $(\mathbf{x}^\top \mathbf{t})^2$ requires only $O(n)$ computations whereas the more direct computation via (13) requires $O(n^2)$.

3.2 Kernel Functions – Implicit Feature Map*

Definition 3.3 (Kernel Function*). Given a feature map ϕ we define its associated **kernel function** $K : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ as

$$K(\mathbf{x}, \mathbf{t}) := \langle \phi(\mathbf{x}), \phi(\mathbf{t}) \rangle, \quad \mathbf{x}, \mathbf{t} \in \mathbb{R}^n \quad (17)$$

Remark 3.4 (Computational Advantage of Kernels). For some feature map ϕ , computing $K(\mathbf{x}, \mathbf{t})$ (Definition 3.3) is independent of N (only dependent on n), whereas computing $\phi(\mathbf{x})$ necessarily depends on N .

Example 3.5 (r -th Order Monomial Kernel). If $\phi(\mathbf{x}) = (x_{i_1}x_{i_2} \dots x_{i_r} : i_1, \dots, i_r \in \{1, \dots, n\})$ then (generalising Example 3.2) we have that

$$K(\mathbf{x}, \mathbf{t}) = (\mathbf{x}^\top \mathbf{t})^r \quad (18)$$

In this case $K(\mathbf{x}, \mathbf{t})$ is computed with $O(n)$ operations, which is essentially independent of r or $N = n^r$. On the other hand, computing $\phi(\mathbf{x})$ requires $O(N)$ operations – Exponential in r !

Question 3.6 (Lower-Order Correlates). So far the feature map has all r -order correlates. How can we change it so that $(r-1)$ -order, $(r-2)$ -order, etc., correlates are included?

3.3 Redundancy of the Feature Map

Warning 3.7 (Feature Map Non-Uniqueness). The feature map is not unique! If ϕ generates K (Definition 3.3) so does $\hat{\phi} = U\phi$ where U is (any!) $N \times N$ orthogonal matrix. Even the dimension of ϕ is not unique!

Proof of Warning 3.7.

$$(U\phi)^\top (U\phi) = \phi^\top U^\top U \phi = \phi^\top \phi \quad (19)$$

□

Example 3.8 (Two Feature Maps for Same Kernel). If $n = 2$, $K(\mathbf{x}, \mathbf{t}) = (\mathbf{x}^\top \mathbf{t})^2$ (cf. Example 3.2) is generated by both

$$\phi(\mathbf{x}) = (x_1^2, x_2^2, x_1x_2, x_2x_1) \quad \text{and} \quad \hat{\phi}(\mathbf{x}) = (x_1^2, x_2^2, \sqrt{2}x_1x_2). \quad (20)$$

3.4 Regularization-based Learning Algorithms^{*}

Let us open a short parenthesis and show that the dual form of ridge regression holds true for other loss functions as well:

$$\mathcal{E}_{\text{emp}_\lambda}(\mathbf{w}) = \sum_{i=1}^m V(y_i, \langle \mathbf{w}, \phi(\mathbf{x}_i) \rangle) + \lambda \langle \mathbf{w}, \mathbf{w} \rangle, \quad \lambda > 0 \quad (21)$$

where $V : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is a loss function.

Theorem 3.9 (Representer Theorem^{*}). *If V is differentiable with respect to its second argument and $\mathbf{w}$ is a minimizer of $\mathcal{E}_\lambda$ (21) then it has the form*

$$\mathbf{w} = \sum_{i=1}^m \alpha_i \phi(\mathbf{x}_i) \quad \Rightarrow \quad f(\mathbf{x}) = \langle \mathbf{w}, \phi(\mathbf{x}) \rangle = \sum_{i=1}^m \alpha_i K(\mathbf{x}_i, \mathbf{x}) \quad (22)$$

where K is the kernel associated with ϕ (Definition 3.3).

Proof of Theorem 3.9. Setting the derivative of $\mathcal{E}_\lambda$ (21) with respect to $\mathbf{w}$ to zero we have

$$\sum_{i=1}^m V'(y_i, \langle \mathbf{w}, \phi(\mathbf{x}_i) \rangle) \phi(\mathbf{x}_i) + 2\lambda \mathbf{w} = \mathbf{0} \quad \Rightarrow \quad \mathbf{w} = \sum_{i=1}^m \alpha_i \phi(\mathbf{x}_i) \quad (23)$$

where V' is the partial derivative of V with respect to its second argument and we defined

$$\alpha_i = -\frac{1}{2\lambda} V'(y_i, \langle \mathbf{w}, \phi(\mathbf{x}_i) \rangle) \quad (24)$$

Thus we conclude that

$$f(\mathbf{x}) = \langle \mathbf{w}, \phi(\mathbf{x}) \rangle = \sum_{i=1}^m \alpha_i \langle \phi(\mathbf{x}_i), \phi(\mathbf{x}) \rangle = \sum_{i=1}^m \alpha_i K(\mathbf{x}_i, \mathbf{x}) \quad (25)$$

□

3.5 Some Remarks

Remark 3.10 (System of Equations for α). Plugging (23) into the right-hand side of (24) we obtain a set of equations for the coefficients α_i :

$$\alpha_i = -\frac{1}{2\lambda} V' \left(y_i, \sum_{j=1}^m K(\mathbf{x}_i, \mathbf{x}_j) \alpha_j \right), \quad i = 1, \dots, m \quad (26)$$

When V is the square loss and $\phi(\mathbf{x}) = \mathbf{x}$ we retrieve the linear equation (6).

Remark 3.11 (Dual Objective Function). Substituting (23) into (21) we obtain an objective function for the α 's:

$$\sum_{i=1}^m V(y_i, (\mathbf{K}\alpha)_i) + \lambda \alpha^\top \mathbf{K} \alpha, \quad \text{where } \mathbf{K} = (K(\mathbf{x}_i, \mathbf{x}_j))_{i,j=1}^m \quad (27)$$

Remark 3.12 (Generality of Representer Theorem). The Representer Theorem (Theorem 3.9) holds true under more general conditions on V (for example V can be any continuous function).

4 What Functions are “Kernels”?*

Question 4.1 (Kernel Characterization). Given a function $K : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$, which properties of K guarantee that there exists a Hilbert space $\mathcal{H}$ (Definition 4.2) and a feature map $\phi : \mathbb{R}^n \rightarrow \mathcal{H}$ such that $K(\mathbf{x}, \mathbf{t}) = \langle \phi(\mathbf{x}), \phi(\mathbf{t}) \rangle$?

Note (Note 1: Infinite-Dimensional Feature Maps). We've generalized the definition of finite-dimensional feature maps $\phi : \mathbb{R}^n \rightarrow \mathbb{R}^N$ to now allow potentially infinite-dimensional feature maps $\phi : \mathbb{R}^n \rightarrow \mathcal{H}$.

Definition 4.2 (Hilbert Space). A **Hilbert space** is an inner product space which also contains the limit points of all its Cauchy sequences (i.e., it is *complete*).

4.1 Positive Semidefinite Kernel

Definition 4.3 (Positive Semidefinite Kernel*). A function $K : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ is **positive semidefinite** if it is symmetric and the matrix $(K(\mathbf{x}_i, \mathbf{x}_j)) : i, j = 1, \dots, k$ is positive semidefinite for every $k \in \mathbb{N}$ and every $\mathbf{x}_1, \dots, \mathbf{x}_k \in \mathbb{R}^n$.

Theorem 4.4 (Kernel–Feature Map Equivalence*). K is positive semidefinite (Definition 4.3) if and only if

$$K(\mathbf{x}, \mathbf{t}) = \langle \phi(\mathbf{x}), \phi(\mathbf{t}) \rangle, \quad \mathbf{x}, \mathbf{t} \in \mathbb{R}^n \quad (28)$$

for some feature map $\phi : \mathbb{R}^n \rightarrow \mathcal{W}$ and Hilbert space $\mathcal{W}$ (Definition 4.2).

Note. We may replace domain $\mathbb{R}^n$ by any abstract set $\mathcal{X}$ in Definition 4.3 and Theorem 4.4.

Proof of Theorem 4.4 (“ $\Leftarrow$ ”). If $K(\mathbf{x}, \mathbf{t}) = \langle \phi(\mathbf{x}), \phi(\mathbf{t}) \rangle$ then we have that

$$\sum_{i,j=1}^m c_i c_j K(\mathbf{x}_i, \mathbf{x}_j) = \left\langle \sum_{i=1}^m c_i \phi(\mathbf{x}_i), \sum_{j=1}^m c_j \phi(\mathbf{x}_j) \right\rangle \quad (29)$$

$$= \left\| \sum_{i=1}^m c_i \phi(\mathbf{x}_i) \right\|^2 \geq 0 \quad (30)$$

for every choice of $m \in \mathbb{N}$, $\mathbf{x}_i \in \mathbb{R}^d$ and $c_i \in \mathbb{R}$, $i = 1, \dots, m$. $\square$

Remark 4.5 (RKHS Construction (“ $\Rightarrow$ ” direction)). The proof of “ $\Rightarrow$ ” in Theorem 4.4 requires the notion of reproducing kernel Hilbert spaces. Informally, one can show that the linear span of the set of functions $\{K(\mathbf{x}, \cdot) : \mathbf{x} \in \mathbb{R}^n\}$ can be made into a Hilbert space $\mathcal{H}_K$ with inner product induced by the definition $\langle K(\mathbf{x}, \cdot), K(\mathbf{t}, \cdot) \rangle_K := K(\mathbf{x}, \mathbf{t})$. In particular, the map $\phi : \mathbb{R}^n \rightarrow \mathcal{H}_K$ defined as $\phi(\mathbf{x}) = K(\mathbf{x}, \cdot)$ is a feature map associated with K . Observe (check!) then with $f(\cdot) := \sum_{i=1}^m \alpha_i K(\mathbf{x}_i, \cdot)$ that $\|f\|^2 = \sum_{i=1}^m \sum_{j=1}^m \alpha_i \alpha_j K(\mathbf{x}_i, \mathbf{x}_j)$.

5 Example Kernels*

5.1 Polynomial Kernel(s)

Proposition 5.1 (Polynomial Kernels*). *If $p : \mathbb{R} \rightarrow \mathbb{R}$ is a polynomial with nonnegative coefficients then*

$$K(\mathbf{x}, \mathbf{t}) = p(\mathbf{x}^\top \mathbf{t}), \quad \mathbf{x}, \mathbf{t} \in \mathbb{R}^n \quad (31)$$

is a positive semidefinite kernel (Definition 4.3).

Example 5.2 (Specific Polynomial Kernels). For example if $a \geq 0$, the following are each positive semidefinite kernels (by Proposition 5.1):

- $K(\mathbf{x}, \mathbf{t}) = (\mathbf{x}^\top \mathbf{t})^r$ (homogeneous polynomial, cf. Example 3.5)
- $K(\mathbf{x}, \mathbf{t}) = (a + \mathbf{x}^\top \mathbf{t})^r$ (inhomogeneous polynomial)
- $K(\mathbf{x}, \mathbf{t}) = \sum_{i=0}^d \frac{a^i}{i!} (\mathbf{x}^\top \mathbf{t})^i$ (truncated exponential)

5.2 Gaussian Kernel

Definition 5.3 (Gaussian Kernel*). An important example of a “radial” kernel is the Gaussian kernel:

$$K(\mathbf{x}, \mathbf{t}) = \exp(-\beta \|\mathbf{x} - \mathbf{t}\|^2), \quad \beta > 0, \quad \mathbf{x}, \mathbf{t} \in \mathbb{R}^n \quad (32)$$

Note. Any corresponding feature map $\phi(\cdot)$ for the Gaussian kernel (Definition 5.3) is ∞ -dimensional.

5.3 Polynomial and Anova Kernel

Definition 5.4 (Anova Kernel*).

$$K_a(\mathbf{x}, \mathbf{t}) = \prod_{i=1}^n (1 + x_i t_i) \quad (33)$$

Compare to the polynomial kernel $K_p(\mathbf{x}, \mathbf{t}) = (1 + \mathbf{x}^\top \mathbf{t})^d$ from Example 5.2.

For $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} :$

$$\phi_p(\mathbf{x}) = \begin{pmatrix} 1 \\ \sqrt{d}x_1 \\ \sqrt{d}x_2 \\ \vdots \\ \sqrt{d}x_n \\ \sqrt{d(d-1)}x_1x_2 \\ \vdots \\ \sqrt{\binom{d}{i_0, i_1, \dots, i_n}} x_1^{i_1} x_2^{i_2} \dots x_n^{i_n} \\ \vdots \end{pmatrix}$$

where $\sum_{j=0}^n i_j = d$.

$$\phi_a(\mathbf{x}) = \begin{pmatrix} 1 \\ x_1 \\ x_2 \\ \vdots \\ x_n \\ x_1x_2 \\ \vdots \\ x_1x_2 \dots x_n \end{pmatrix}$$

Exercise 5.5 (Anova Feature Map Verification). Argue that $\langle \phi_a(\mathbf{x}), \phi_a(\mathbf{t}) \rangle = K_a(\mathbf{x}, \mathbf{t})$ (Definition 5.4).

6 Kernel Construction*

Which operations/combinations (e.g., products, sums, composition, etc.) of a given set of kernels is still a kernel (Definition 4.3)?

If we address this question we can build more interesting kernels starting from simple ones.

Example 6.1 (Motivating Question). We have already seen that $K(\mathbf{x}, \mathbf{t}) = (\mathbf{x}^\top \mathbf{t})^r$ is a kernel (Example 3.5). For which class of functions $p : \mathbb{R} \rightarrow \mathbb{R}$ is $p(\mathbf{x}^\top \mathbf{t})$ a kernel? More generally, if K is a kernel when is $p(K(\mathbf{x}, \mathbf{t}))$ a kernel?

6.1 General Linear Kernel

Proposition 6.2 (General Linear Kernel). *If $\mathbf{A}$ is an $n \times n$ positive semidefinite matrix the function $K : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ defined by*

$$K(\mathbf{x}, \mathbf{t}) = \mathbf{x}^\top \mathbf{A} \mathbf{t} \quad (34)$$

is a kernel.

Proof of Proposition 6.2. Since $\mathbf{A}$ is positive semidefinite we can write it in the form $\mathbf{A} = \mathbf{R}\mathbf{R}^\top$ for some $n \times n$ matrix $\mathbf{R}$. Thus K is represented by the feature map $\phi(\mathbf{x}) = \mathbf{R}^\top \mathbf{x}$ (cf. Definition 3.3).

Alternatively, note that:

$$\sum_{i,j} c_i c_j \mathbf{x}_i^\top \mathbf{A} \mathbf{x}_j = \sum_{i,j} c_i c_j (\mathbf{R}^\top \mathbf{x}_i)^\top (\mathbf{R}^\top \mathbf{x}_j) \quad (35)$$

$$= \sum_i c_i [(\mathbf{R}^\top \mathbf{x}_i)]^\top \left[\sum_j c_j (\mathbf{R}^\top \mathbf{x}_j) \right] \quad (36)$$

$$= \left\| \sum_i c_i \mathbf{R}^\top \mathbf{x}_i \right\|^2 \geq 0 \quad (37)$$

□

6.2 Kernel Composition

Proposition 6.3 (Kernel Composition). *More generally, if $K : \mathbb{R}^N \times \mathbb{R}^N \rightarrow \mathbb{R}$ is a kernel (Definition 4.3) and $\phi : \mathbb{R}^n \rightarrow \mathbb{R}^N$, then*

$$\tilde{K}(\mathbf{x}, \mathbf{t}) = K(\phi(\mathbf{x}), \phi(\mathbf{t})) \quad (38)$$

is a kernel from $\mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$.

Proof of Proposition 6.3. By hypothesis, K is a kernel and so, for every $\mathbf{x}_1, \dots, \mathbf{x}_m \in \mathbb{R}^n$ the matrix $(K(\phi(\mathbf{x}_i), \phi(\mathbf{x}_j))) : i, j = 1, \dots, m$ is positive semidefinite (Definition 4.3). □

Remark 6.4 (Special Case). In particular, Proposition 6.2 corresponds to Proposition 6.3 with $K(\mathbf{x}, \mathbf{t}) = \mathbf{x}^\top \mathbf{t}$ and $\phi(\mathbf{x}) = \mathbf{R}^\top \mathbf{x}$.

6.3 Kernel Construction (Continued)

Question 6.5 (General Kernel Construction). If $K_1, \dots, K_q$ are kernels on $\mathbb{R}^n$ and $F : \mathbb{R}^q \rightarrow \mathbb{R}$, when is the function

$$F(K_1(\mathbf{x}, \mathbf{t}), \dots, K_q(\mathbf{x}, \mathbf{t})), \quad \mathbf{x}, \mathbf{t} \in \mathbb{R}^n \quad (39)$$

a kernel?

Equivalently: when for every choice of $m \in \mathbb{N}$ and $\mathbf{A}_1, \dots, \mathbf{A}_q$ $m \times m$ positive semidefinite matrices, is the following matrix positive semidefinite?

$$(F(A_{1,ij}, \dots, A_{q,ij}) : i, j = 1, \dots, m) \quad (40)$$

We discuss some examples of functions F for which the answer to Question 6.5 is YES.

6.4 Nonnegative Combination of Kernels

Proposition 6.6 (Nonnegative Combination*). *If $\lambda_j \geq 0$, $j = 1, \dots, q$ then $\sum_{j=1}^q \lambda_j K_j$ is a kernel.*

This fact is immediate (a non-negative combination of positive semidefinite matrices is still positive semidefinite).

Example 6.7 (Component Kernels). Let $q = n$ and $K_j(\mathbf{x}, \mathbf{t}) = x_j t_j$.

Corollary 6.8 (Scaling and Addition). *From Proposition 6.6:*

- aK_1 is a kernel if $a \geq 0$
- $K_1 + K_2$ is a kernel

6.5 Product of Kernels

Proposition 6.9 (Schur Product of Kernels*). *The pointwise product of two kernels K_1 and K_2*

$$K(\mathbf{x}, \mathbf{t}) := K_1(\mathbf{x}, \mathbf{t})K_2(\mathbf{x}, \mathbf{t}), \quad \mathbf{x}, \mathbf{t} \in \mathbb{R}^d \quad (41)$$

is a kernel.

Proof of Proposition 6.9. Idea: The fact that the element-wise product of positive semidefinite matrices is again positive semidefinite (the **Schur product theorem**) implies that product of kernels is again a kernel.

Thus we need to show that if $\mathbf{A}$ and $\mathbf{B}$ are positive semidefinite matrices, so is $\mathbf{C} = (A_{ij}B_{ij} : i, j = 1, \dots, n)$ ($\mathbf{C}$ is also called the Schur product of $\mathbf{A}$ and $\mathbf{B}$).

Since $\mathbf{A}$ and $\mathbf{B}$ are positive semidefinite we can write them in the form $\mathbf{A} = \mathbf{U}\mathbf{U}^\top$ and $\mathbf{B} = \mathbf{V}\mathbf{V}^\top$ for some $n \times n$ matrices $\mathbf{U}$ and $\mathbf{V}$.

$$\sum_{i,j=1}^m z_i z_j C_{ij} = \sum_{i,j} z_i z_j \sum_r U_{ir} U_{jr} \sum_s V_{is} V_{js} \quad (42)$$

$$= \sum_{i,j} \sum_{rs} z_i z_j U_{ir} U_{jr} V_{is} V_{js} \quad (43)$$

$$= \sum_{rs} \sum_{ij} z_i z_j U_{ir} U_{jr} V_{is} V_{js} \quad (44)$$

$$= \sum_{rs} \sum_i z_i U_{ir} V_{is} \sum_j z_j U_{jr} V_{js} \quad (45)$$

$$= \sum_{rs} \left(\sum_i z_i U_{ir} V_{is} \right)^2 \geq 0 \quad (46)$$

□

6.6 Summary of Kernel Constructions

Theorem 6.10 (Kernel Closure Properties*). *If K_1, K_2 are kernels (Definition 4.3), $a \geq 0$, $\mathbf{A}$ is a symmetric positive semi-definite matrix, K a kernel on $\mathbb{R}^N$ and $\phi : \mathbb{R}^n \rightarrow \mathbb{R}^N$ then the following functions are positive semidefinite kernels on $\mathbb{R}^n$:*

1. $\mathbf{x}^\top \mathbf{A} \mathbf{t}$ (Proposition 6.2)
2. $K_1(\mathbf{x}, \mathbf{t}) + K_2(\mathbf{x}, \mathbf{t})$ (Corollary 6.8)
3. $aK_1(\mathbf{x}, \mathbf{t})$ (Corollary 6.8)
4. $K_1(\mathbf{x}, \mathbf{t})K_2(\mathbf{x}, \mathbf{t})$ (Proposition 6.9)
5. $K(\phi(\mathbf{x}), \phi(\mathbf{t}))$ (Proposition 6.3)

6.7 Polynomial of Kernels

Corollary 6.11 (Polynomial of Kernels*). *Let $F = p$ where $p : \mathbb{R}^q \rightarrow \mathbb{R}$ is a polynomial in q variables with nonnegative coefficients. By properties 2, 3 and 4 of Theorem 6.10 we conclude that p is a valid function.*

In particular if $q = 1$,

$$\sum_{i=1}^d a_i (K(\mathbf{x}, \mathbf{t}))^i \quad (47)$$

is a kernel if $a_1, \dots, a_d \geq 0$.

6.8 Polynomial Kernels

The above observation (Corollary 6.11) implies that if $p : \mathbb{R} \rightarrow \mathbb{R}$ is a polynomial with nonnegative coefficients then $p(\mathbf{x}^\top \mathbf{t})$, $\mathbf{x}, \mathbf{t} \in \mathbb{R}^n$ is a kernel on $\mathbb{R}^n$ (cf. Proposition 5.1). In particular if $a \geq 0$ the following are valid polynomial kernels:

- $(\mathbf{x}^\top \mathbf{t})^r$
- $(a + \mathbf{x}^\top \mathbf{t})^r$
- $\sum_{i=0}^d \frac{a^i}{i!} (\mathbf{x}^\top \mathbf{t})^i$

6.9 ‘Infinite Polynomial’ Kernel

Proposition 6.12 (Exponential Kernel). *If in the last equation we set $r = \infty$ the series*

$$\sum_{i=0}^r \frac{a^i}{i!} (\mathbf{x}^\top \mathbf{t})^i \quad (48)$$

converges everywhere uniformly to $\exp(a\mathbf{x}^\top \mathbf{t})$ showing that this function is also a kernel (as a limit of kernels from Corollary 6.11).

Assume for simplicity that $n = 1$. A feature map corresponding to the kernel $\exp(axt)$ is

$$\phi(x) = \left(1, \sqrt{ax}, \sqrt{\frac{a^2}{2}}x^2, \sqrt{\frac{a^3}{6}}x^3, \dots \right) = \left(\sqrt{\frac{a^i}{i!}}x^i : i \in \mathbb{N} \right) \quad (49)$$

Remark 6.13 (Infinite-Dimensional Feature Space). The feature space corresponding to (49) has an infinite dimensionality!

6.10 Translation Invariant and Radial Kernels

Definition 6.14 (Translation Invariant and Radial Kernels). We say that a kernel $K : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ is

- **Translation invariant** if it has the form

$$K(\mathbf{x}, \mathbf{t}) = H(\mathbf{x} - \mathbf{t}), \quad \mathbf{x}, \mathbf{t} \in \mathbb{R}^d \quad (50)$$

where $H : \mathbb{R}^d \rightarrow \mathbb{R}$ is a differentiable function.

- **Radial** if it has the form

$$K(\mathbf{x}, \mathbf{t}) = h(\|\mathbf{x} - \mathbf{t}\|), \quad \mathbf{x}, \mathbf{t} \in \mathbb{R}^d \quad (51)$$

where $h : [0, \infty) \rightarrow [0, \infty)$ is a differentiable function.

6.11 The Gaussian Kernel

An important example of a radial kernel (Definition 6.14) is the Gaussian kernel (Definition 5.3):

$$K(\mathbf{x}, \mathbf{t}) = \exp(-\beta \|\mathbf{x} - \mathbf{t}\|^2), \quad \beta > 0, \mathbf{x}, \mathbf{t} \in \mathbb{R}^d \quad (52)$$

Proposition 6.15 (Gaussian Kernel is PSD*). *The Gaussian kernel (52) is a valid kernel because it is the product of two kernels (by Proposition 6.9):*

$$K(\mathbf{x}, \mathbf{t}) = \underbrace{\exp(-\beta(\mathbf{x}^\top \mathbf{x} + \mathbf{t}^\top \mathbf{t}))}_{\text{kernel with } \phi(\mathbf{x}) = \exp(-\beta \mathbf{x}^\top \mathbf{x})} \cdot \underbrace{\exp(2\beta \mathbf{x}^\top \mathbf{t})}_{\text{kernel by Proposition 6.12}} \quad (53)$$

Exercise 6.16 (Gaussian Feature Map). Can you find a feature map representation for the Gaussian kernel (Definition 5.3)?

Remark 6.17 (Kernels on General Domains). Defining useful kernels on new domains $\mathcal{X}$ allows for a host of ML algorithms to be transferred to that domain. For example ridge regression, k -NN, SVMs, k -means, PCA, etc.

7 Computational Summary

7.1 Summary: Computation with Basis Functions

Data: $X, (m \times n); \mathbf{y}, (m \times 1)$

Basis Functions: $\phi_1, \dots, \phi_N$ where $\phi_i : \mathbb{R}^n \rightarrow \mathbb{R}$

Feature Map: $\phi : \mathbb{R}^n \rightarrow \mathbb{R}^N$ (Definition 3.3 gives the kernel from the feature map)

$$\phi(\mathbf{x}) = (\phi_1(\mathbf{x}), \dots, \phi_N(\mathbf{x})), \quad \mathbf{x} \in \mathbb{R}^n \quad (54)$$

Mapped Data Matrix:

$$\Phi := \begin{pmatrix} \phi(\mathbf{x}_1) \\ \vdots \\ \phi(\mathbf{x}_m) \end{pmatrix} = \begin{pmatrix} \phi_1(\mathbf{x}_1) & \cdots & \phi_N(\mathbf{x}_1) \\ \vdots & \ddots & \vdots \\ \phi_1(\mathbf{x}_m) & \cdots & \phi_N(\mathbf{x}_m) \end{pmatrix}, \quad (m \times N) \quad (55)$$

Regression Coefficients: $\mathbf{w} = (\Phi^\top \Phi + \lambda I_N)^{-1} \Phi^\top \mathbf{y}$

Regression Function: $\hat{y}(\mathbf{x}) = \sum_{i=1}^N w_i \phi_i(\mathbf{x})$

7.2 Summary: Computation with Kernels

Data: $X, (m \times n); \mathbf{y}, (m \times 1)$

Kernel Function: $K : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ (Definition 3.3)

Kernel Matrix:

$$\mathbf{K} := \begin{pmatrix} K(\mathbf{x}_1, \mathbf{x}_1) & \cdots & K(\mathbf{x}_1, \mathbf{x}_m) \\ \vdots & \ddots & \vdots \\ K(\mathbf{x}_m, \mathbf{x}_1) & \cdots & K(\mathbf{x}_m, \mathbf{x}_m) \end{pmatrix}, \quad (m \times m) \quad (56)$$

Regression Coefficients: $\boldsymbol{\alpha} = (\mathbf{K} + \lambda I_m)^{-1} \mathbf{y}$

Regression Function: $\hat{y}(\mathbf{x}) = \sum_{i=1}^m \alpha_i K(\mathbf{x}_i, \mathbf{x})$ (the dual form from Theorem 3.9)

8 Problems

8.1 Problems – 1

1. Prove results on page 9.
2. Consider the solution to linear regression optimisation problem. When is it advantageous to compute it via the primal solution (Remark 2.1)? When is it advantageous to compute it via the dual solution? Explain why.
3. Given a kernel $K : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ where $K(\mathbf{x}, \mathbf{t}) := (1 + \langle \mathbf{x}, \mathbf{t} \rangle)^2$. Find a feature map $\phi : \mathbb{R}^2 \rightarrow \mathbb{R}^6$ which corresponds to the kernel (cf. Proposition 5.1).
4. For each of the following functions $K : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ argue whether they are a valid kernel (Definition 4.3) and when the answer is positive derive an associated feature map representation.

(a) $K(\mathbf{x}, \mathbf{t}) = \mathbf{x}^\top D \mathbf{t}$, where D is the matrix $\begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix}$ (cf. Proposition 6.2)

(b) $K(\mathbf{x}, \mathbf{t}) = \mathbf{x}^\top D \mathbf{t}$, where D is the matrix $\begin{bmatrix} -1 & 2 \\ 2 & 4 \end{bmatrix}$

(c) $K(\mathbf{x}, \mathbf{t}) = \exp(x_1 t_1)$, where x_1 is the first component of the vector $\mathbf{x}$ and, likewise, t_1 is the first component of the vector $\mathbf{t}$ (cf. Proposition 6.12)

(d) $K(\mathbf{x}, \mathbf{t}) = \mathbf{x}^\top \mathbf{t} - (\mathbf{x}^\top \mathbf{t})^2$.

(e) Now prove that if $K : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ is a given valid kernel then the following transformed kernels are also valid:

- i. $K(A\mathbf{x}, A\mathbf{t})$, where A is a given 2×2 matrix (cf. Proposition 6.3).
 - ii. $f(\mathbf{x})K(\mathbf{x}, \mathbf{t})f(\mathbf{t})$, where f is a given real-valued function.
5. Consider a Gaussian kernel function $K : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ defined by $K(\mathbf{x}, \mathbf{z}) := e^{-\|\mathbf{x} - \mathbf{z}\|^2}$ (Definition 5.3 with $\beta = 1$), does there exist a finite-dimensional feature map representation? I.e., does there exist a $\phi : \mathbb{R}^n \rightarrow \mathbb{R}^d$ such that $K(\mathbf{x}, \mathbf{z}) = \langle \phi(\mathbf{x}), \phi(\mathbf{z}) \rangle$? Indicate an answer “yes” or “no” and provide an argument supporting your answer (cf. Theorem 4.4). [hard]